ML lab 3

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Class:- C3

Batch:- C3

Roll no:- 44

```
import pandas as pd
import numpy as np
df = pd.read_csv("enjoy.csv")
print(df_csv.head())
```

```
     Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool   Change        Yes
4  Sunny    Cold   Normal    Weak  Warm     Same         No
```

```
X=df.iloc[:,:-1].values
y=df.iloc[:,-1].values
```

```
import numpy as np

def find_s(X, y):
    hypothesis = None
    print("initial hypothesis:", hypothesis)

    for i in range(len(X)):
        if y[i] == "Yes":
            if hypothesis is None:
                hypothesis = X[i].copy()
            else:
                for j in range(len(hypothesis)):
                    if hypothesis[j] == X[i][j]:
                        hypothesis[j] = "?"

            print(f"\nAfter Training Example {i+1}")
            print(hypothesis)

    return hypothesis

final_hypothesis = find_s(X, y)
print("\nfinal hypothesis")
print(final_hypothesis)
```

```
initial hypothesis: None

After Training Example 1
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

After Training Example 2
['?' '?' 'Normal' '?' '?' '?']

After Training Example 4
['?' '?' 'Normal' '?' '?' '?']

After Training Example 7
['?' '?' '?' '?' '?' '?']

After Training Example 9
['?' '?' '?' '?' '?' '?']

After Training Example 10
['?' '?' '?' '?' '?' '?']

final hypothesis
['?' '?' '?' '?' '?' '?']
```

```
import copy

def candidate_elimination(concept, target):
    specific = concept[0].copy()
    general = [["?" for i in range(len(specific))]]

    print("\ninitial Sc:", specific)
    print("\ninitial G:", general)

    for i, h in enumerate(concept):
        if target[i] == "Yes":
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    specific[x] = "?"
        else:
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    general.append(["?" if j != x else specific[x] for j in range(len(specific))])

        print("\nExample", i + 1)
        print("S=", specific)
        print("G=", general)

    return specific, general


S, G = candidate_elimination(X, y)

print("\nfinal specific Boundary")
print(S)

print("\nfinal general Boundary")

for g in G:
    print(g)
```

```
initial Sc: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

initial G: [['?', '?', '?', '?', '?', '?']]

Example 1
S= ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
G= [['?', '?', '?', '?', '?', '?']]

Example 2
S= ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
G= [['?', '?', '?', '?', '?', '?']]

Example 3
S= ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 4
S= ['Sunny' 'Warm' '?' 'Strong' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 5
S= ['Sunny' 'Warm' '?' 'Strong' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 6
S= ['Sunny' 'Warm' '?' 'Strong' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 7
S= ['Sunny' 'Warm' '?' '?' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 8
S= ['Sunny' 'Warm' '?' '?' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 9
S= ['Sunny' 'Warm' '?' '?' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'

Example 10
S= ['Sunny' 'Warm' '?' '?' '?' '?']
G= [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?'
```

```
final specific Boundary
['Sunny' 'Warm' '?' '?' '?' '?']

final general Boundary
['?', '?', '?', '?', '?', '?']
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
```

Start coding or generate with AI.